Zili Xie

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EDUCATION

New York University, Brooklyn, NY **5/ 2021**
Master of Science in Computer Science
University of Toronto, St. George, Toronto, ON, Canada **5/ 2019**
Bachelor of Science with Distinction | Specialist in Computer Science & Minor in Statistics | Rank Top 25% in CS Department

TECHNICAL SKILLS

Languages: C (kernel, socket), C#, C++, Python (Anaconda), Java, SQL, XQuery, Linux shell, HTML, CSS, Verilog, Assembly, Javascript, R language, Powershell, Racket, Haskell, DDL, GLSL, PHP.
Develop Tools: CMake, Clion, Jupyter Notebook, Colab, Pyzo, Visual Studio, Eclipse, Android Studio, R Studio.
AI & ML: **(library):** Pytorch, numpy, Autograd. **(model):** CNN, RNN, attention-based Machine Translation.
Vision/Graphic: **(library):** OpenCV, OpenGL, Eigen, Libigl. **(algo):** Blinn-Phong reflection, Catmull-Clark subdiv. **(model):** Ray Tracing, AABB Tree, Object Texture Mapping, Shader Pipeline.
Database: SQLite, MySQL, Postgresql, MongoDB.
Web Develop: Node js (express, ejs, body-parser), Bootstrap, Angular, React js, jQuery, REST.
Version Control & Automation: Git, SVN, Jira, Jenkins, IBM Urbancode.

WORK & RESEARCH EXPERIENCES

System Analyst, Solution Engineer, DevOps, Ontario Teachers' Pension Plan, Toronto **5/ 2017 - 4/ 2018**
- Created automated build jobs in Jenkins and deploy processes in IBM Urbancode for Windows applications ran by OneRisk team and PM/ER team. Created application templates in Jenkins and Urbancode for the DevOps team.
- Designed a parallel system that integrates all build jobs, deploy processes and SonarQube Vulnerabilities analyses. The new system makes the continuous integration workflow 50% faster than before.
- Developed an ASP.NET Web application that allows OTPP Github users to quickly obtain information about specific Github Repos, and get release zip file based on repo name, tag or release version.

Programmer Assistant, China Mobile Branch, TERADATA, Guangzhou **5/ 2016 - 8/ 2016**
- Wrote Teradata SQL for clients, also created SQL templates for company's different projects.
- Developed Linux scripts that imports data from excel documents to TERADATA warehouse.

Research Assistant, The Identification of Running Mode of Variable Frequency Air-condition, Guangzhou **8/ 2018**
- Participated in a research investigating the running mode of Air-condition held by School of Mechanical Engineering at SCUT.
- Applied methods of clustering algorithm, neutral network models, etc to analyze data collected from three different operational modes, based on which put forward suggestions on optimizing the control of the variable frequency air condition.

PROJECTS

CNN for Image Colorization (python3, numpy, pytorch, colab) **3/ 2019**
- Implemented and trained a CNN model that takes grey-scale image as input and produces colorized image as output. The model frames colorization as a classification problem and uses UNet to improve performance.

Region Filling and Object Removal by Image Inpainting (python2, numpy, OpenCV, Kivy) **3/ 2019**
- Implemented an Image inpainting tool that can remove any objects in image and refill the unknown region using Criminisi's *Exemplar-Based Image Inpainting* technique.

Neural Network Language Model (python3, numpy, pytorch) **2/ 2019**
- Implemented and trained a neural network that takes word sequences as input and learns to predict word comes next.

Bounding Volume Hierarchy (C++, OpenGL, Libigl, AABB tree) **12/ 2018**
- Built a program that encloses groups of objects (e.g., points, triangle meshes, bounding boxes) in scene into an Axis-Aligned Bounding-Box (AABB) tree, reducing the computational cost of ray-intersection queries, distance queries and objects intersection queries to *O(nlogn)*.

Shader Pipeline (C++, glsl, OpenGL, Libigl) **12/ 2018**
- Implemented a real-time rendering shader pipeline using the OpenGL Shading Language (glsl) that renders an animated "Earth-Moon" system with bumpy surface and textures on planets produced by *Perlin* Noise.

Ray Tracer (C++, OpenGL, Reflection Model) **11/ 2018**
- Developed a raytracing program that produces accurate scenes illumination using the GLlibrary and Blinn-Phong reflection model.

Pacman Game AI (python3) **10/ 2018**
- Implemented a Pacman agent that eats all the dots in a maze before timeout and avoids being caught by enemy ghosts. The Pacman's AI is implemented using A* search, Minimax and Expectimax game tree, Alpha-Beta pruning technique.

Simple Router (C, IP/TCP/UDP/ARP/ICMP, Mininet, VirtualBox) **12/ 2017**
- Built a router in C that parses the received Ethernet frames and forwards the IP packet to correct outgoing interface. It also sends ICMP in response to the sending host and uses ARP request to determine the next-hop MAC address.

Web Application: Designforum (Javascript, NodeJS, Express, JQuery, CSS, EJS) **12/ 2016**
- Developed a web application for designers to display their products of art, design, or creative ideas using Node.js to server part, EJS to make front-end and MongoDB for data storage.
- Third-part user authentication: Login using Facebook account without signing up in the form.
- Administrative View: Admin users have their own view that allow them to view and edit all other user profiles.

"FlightQuery" Android Application (Java, Android Studio) **6/ 2016**
- Built a real time flight schedule inquiry app with autosuggestions on itinerary and autocompletion on name of places.